A Retrospective Analysis of Internet Crimes Against Children Across U.S. Task Forces, 2010–2026

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Abstract Submission — *Internet Histories* Special Issue: Interrogating Trust & Safety
Written April 2026

Abstract

The thirty-year arc from the public web's emergence to today's AI-saturated social media landscape has transformed almost every aspect of humanity. From how we communicate, consume content, and conduct almost every aspect of our lives, the internet and computers have revolutionized what human beings are capable of. The darker corollary to that revolution — underrepresented, underexamined, and rarely centered in the history of the internet — is what that same infrastructure made possible for those who exploit children. This paper traces that transformation through a novel lens: analysis of publicly available case reports from the Internet Crimes Against Children (ICAC) Task Force Program—a federally coordinated network of 61 task forces representing over 5,400 agencies—analyzed at scale for the first time using an open-source retrospective case analysis system. Rather than analyzing ICAC history and policy chronologically, this research performs extensive case studies and retrospective analysis over publicly available ICAC cases: a longitudinal archive of how internet-facilitated child exploitation has actually manifested in investigated cases from 2010 to 2026, and how the law enforcement and trust and safety ecosystem has responded, adapted, and strained under the weight of each successive technological wave.

The contribution of this paper is both historical and methodological. Historically, we present 100 structured case studies drawn from the CaseLinker dataset—spanning platform involvement, victim and perpetrator demographics, investigation type (proactive, undercover, online, vs. reactive), geographic and jurisdictional patterns, and agency coordination networks. These case studies are organized across four eras corresponding to distinct technological inflection points: the social platform era (2010–2014), the mobile-first transition (2015–2018), Platform proliferation and expansion (2019–2022), and the recent social media and generative AI period (2023–2026). Each era is characterized by shifting perpetrator methodologies, evolving platform surfaces, and corresponding adaptations in law enforcement response—patterns that are invisible at the individual case level and only emerge through systematic cross-case analysis. We situate these findings within the broader Trust and Safety literature,

treating ICAC task forces as an example of institutional response to digital harm—one that has become burdened by the full weight of the internet's expansion — 20.5 million CyberTip reports in 2024 alone — and whose operational history offers grounded insight into how emerging technologies have compounded harm against vulnerable populations and explores how understanding that history can better equip those working to protect children today.

Research Methodology

This research is grounded in CaseLinker, an open-source retrospective case analysis system developed to process, structure, and surface patterns from publicly available ICAC press releases and task force case reports. The system is designed to be auditable, interpretable, and appropriate for the sensitivity of the subject matter—operating exclusively on already-public, already-redacted case summaries and producing no outputs that identify individuals.

Ingestion and normalization. The ingestion pipeline accepts PDF case reports from individual ICAC task forces, each of which publishes in a distinct format. A per-source extraction layer handles case batching and text normalization, formatting PDFs with case narratives before downstream processing.

Feature extraction. Extraction operates in two layers. The primary layer is a deterministic pattern extraction system: a domain calibrated set of regular expressions and linguistic rules that extract structured fields from case text—offense type, platform mentioned, victim age range, perpetrator–victim relationship, investigation classification (proactive vs. reactive), prosecutorial outcome, and agency involvement, among others. Every extracted feature is traceable to its source text in the database, enabling full auditability. The secondary layer is a Named Entity Recognition system (Stanford Stanza) that augments pattern extraction with semantic entity detection—particularly for law enforcement organization identification, where surface-form variation across agencies makes rule-based approaches brittle. NER outputs are merged with pattern outputs under a precedence system that preserves deterministic primacy; the machine learning layer is additive and optional, not load-bearing.

Analysis stack. Case-level features feed a multi-component analysis stack. A 9 dimension facet tree supports cohort-level navigation and constraint-based case filtering—enabling structured queries such as grooming-platform cases involving victims under 14 in reactive investigations from 2018–2022, returned as a structured case cohort with membership counts and facet paths. A Jaccard-similarity clustering layer groups cases by feature overlap, surfacing recurring case profiles that cut across jurisdictions and years. A rule-based triage system with documented, inspectable scoring weights assigns priority tiers to cases based on severity indicators, and a Random Forest classifier trained on triage labels supports further analysis –with outputs explicitly downstream of and accountable to the rule layer.

Case study selection. The 100 case studies will be selected using stratified sampling across the four historical eras and five analytic dimensions—platform, demographics, investigation type, geography, and agency network—to ensure coverage of both dominant patterns and structurally important edge cases. Each case study is written as a structured narrative: platform context, perpetrator methodology, investigative approach, prosecutorial outcome, and relationship to the broader technological moment of its era. The goal is not statistical generalization from the sample, but grounded historical illustration of patterns that aggregate analysis surfaces and that individual case reading cannot reveal. All case data is drawn from public sources. The full source corpus, extraction code, and analysis pipeline are released as open-source software to support replication and extension by other researchers.

CaseLinker currently contains 530 cases, 5516 extracted features, and 6 sources. The search, triage, analysis, clustering, and audit features have been implemented and can be used to aid with the case studies. This project received a determination from the University of Massachusetts Amherst Human Research Protection Office (HRPO Determination #7668); this research does not contain private or identifiable information under federal regulations [45 CFR 46.102(f)(1), (2)]

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